VIGIL: Validity-Gated Retrieval for Product–Litigation Risk Entailment

Anonymous ACL submission

Abstract

Retrieval-augmented generation (RAG) treats semantically similar documents as interchangeable evidence, but in law a lexically close precedent may already have been narrowed, superseded, or overruled at the date that matters. We study this failure in a product-facing setting and introduce *Product–Litigation Risk Entailment* (PLRE), a directional task that asks whether an AI product's attributes entail exposure to a legal-risk theory under precedent that is valid at a given query date. Our system, VIGIL (Validity-Interval Gating in Litigation), infers *when* each authority stops being operative: it extracts treatment edges (overrules, supersedes, narrows) from later opinions, turns them into validity intervals, and validates them against gold annotations and a commercial citator (negative-treatment micro-F1 0.81). A deliberately simple retrieval-time gate applies these intervals before generation, and a goal-conditioned agentic loop stops when evidence coverage, not legal correctness, is satisfied. On VIGIL-Dataset (1,247 U.S. AI-litigation cases, 312 product-mapping cases), validity gating raises PLRE F1 from 0.72 to 0.78 (paired ΔF1 95% CI $[0.03, 0.10]$) and cuts stale citations relative to flat similarity; on a human-adjudicated triage subset (n=82) VIGIL reaches macro F1 0.79 vs. 0.75 for the strongest baseline, with a modest paired gap whose confidence interval includes zero. We release the dataset with per-record label provenance, annotation guidelines, and evaluation scripts. VIGIL is decision support for qualified reviewers, not legal advice.

1 Introduction

Generative-AI products increasingly face litigation over training data, output reproduction, model behavior, and downstream misuse. For the engineering organizations that build them, the operational question is not merely whether a lawsuit exists, but whether a theory of liability maps to a specific product attribute, whether the supporting authority was valid at the relevant date, and which mitigation should be reviewed by counsel. Keyword alerts surface noisy filings, docket trackers rarely map cases to product features, and conventional retrieval-augmented generation (RAG) (Lewis et al., 2020) treats semantically similar legal documents as comparable even when later decisions have narrowed or superseded them.

We formalize this need as *Product–Litigation Risk Entailment* (PLRE): deciding whether product attributes entail exposure to a legal-risk theory under time-valid precedent. PLRE differs from standard legal textual entailment (Rabelo et al., 2022) in three ways: the premise is an engineering description rather than legal text, the relation is directional (product attribute $\rightarrow$ risk theory), and the support set is conditioned on which authority was operative at the query date. Unlike commercial citators such as Shepard's and KeyCite, we build a public-corpus, automatically extracted, time-indexed treatment layer and connect it to product-facing entailment.

We present VIGIL (**Validity-Interval Gating in Litigation**), a continuous sensing system built around this task. Its core component is a supersession-aware treatment layer: it infers when each authority stops being operative from treatment edges in *later* opinions, and a simple gate (Eq. (3)) applies this before generation, rather than flagging stale citations after an LLM has already produced a report. Around this layer, a goal-conditioned agentic loop exposes unmet evidence requirements and stops when calibrated evidence coverage is satisfied. The layer's form is not specific to law—it is defined for any time-stamped corpus in which later documents can supersede earlier ones—but we evaluate it only for law, within VIGIL.

Our contributions are:

- a supersession-aware treatment layer: treatment edges extracted from public opinions, validated

against gold spans and a commercial citator, and propagated into validity intervals that gate retrieval at query time (§4.4);

- PLRE, an initial product-conditioned legal-risk entailment benchmark with non-VIGIL baselines and ablations that separate temporal validity from flat similarity (§4.5, §6);
- a goal-conditioned agentic loop whose stopping rule controls evidence coverage, with a conformal omission-risk guarantee (§3.1);
- VIGIL-Dataset, with per-record label provenance, annotation guidelines, and a human-adjudicated evaluation subset (§5), plus a 30-day live deployment (Appendix D), and a public, licence-free testbed for supersession-aware retrieval built from overruled U.S. Supreme Court precedent (Appendix I.1).

2 Background and Scope

Legal-risk monitoring for AI products. Copyright litigation against generative-AI systems has grown sharply since 2022, challenging web-scale training data (Henderson et al., 2023; Levendowski, 2018) under theories spanning DMCA §1202 information removal, direct infringement, and vicarious liability for outputs (Grimmelmann, 2016). Legal-operations teams track dockets, news, and internal product changes, but the bridge from a litigation event to a product-specific engineering action is usually manual. VIGIL targets this product-facing layer.

3 Problem Formulation

Let $\mathcal{D} = \{d_1, \ldots, d_n\}$ be public litigation documents with metadata and text, and let P be a product specification describing architecture, data provenance, modality, and deployment context. VIGIL returns a structured risk report R^* and mitigation candidates G^* that must satisfy an auditable goal predicate $\mathcal{G}$ (§3.1). Report quality is not optimized directly; we evaluate it by severity-label F1 (RQ1), blinded report ratings (RQ2), and the coverage loss L_τ (RQ6). The core difficulty is a domain gap: product specifications use engineering terms (data pipelines, API distribution), whereas litigation documents use causes of action, procedural posture, and precedent. PLRE (§4.5) therefore requires mapping engineering-language attributes to legal-risk theories while conditioning the support set on authority valid at the query date.

3.1 Goal Predicate and Stopping Rule

VIGIL treats report generation as goal-conditioned planning. The predicate is:

$$\mathcal{G}(R) = \mathcal{G}_{\text{risk}}(R) \wedge \mathcal{G}_{\text{evidence}}(R) \wedge \mathcal{G}_{\text{guide}}(R), \quad (1)$$

where $\mathcal{G}_{\text{risk}}$ requires every risk item to have a severity label, $\mathcal{G}_{\text{evidence}}$ requires at least k supporting documents above relevance threshold τ, and $\mathcal{G}_{\text{guide}}$ requires mitigation text for each High or Medium item. The controller chooses among SEARCH, ANALYZE, SUMMARIZE, MAP, VERIFY, and REPORT; the deterministic (LLM-free) Goal Checker either terminates, emits a structured critique to close residual gaps, or escalates unmet conditions. For example, if a High-risk item has only one valid authority after gating, the critique records the missing theory, product attribute, jurisdiction, query date, and required evidence count; the Planner rewrites this into narrower precedent and entity queries rather than simply retrying the same prompt.

Let $\Phi(R_t)$ count the unmet atomic sub-conditions of $\mathcal{G}$, let $\mathcal{T}_c^{\text{H}}$ be the human-labeled High-risk theories of case c that are *retrievable*, i.e., supported by at least one authority in $\mathcal{D}$ that is valid at t_q, and define the coverage loss at evidence threshold τ as

$$L_\tau(c) = \mathbf{1}\big[\exists\, T \in \mathcal{T}_c^{\text{H}} :\ T \notin R^*\big]. \quad (2)$$

Proposition 1 (Coverage-controlled stopping with escalation)**.** *Under monotone evidence caching and attainable progress, $\Phi(R_t)$ is non-increasing, reaches zero in at most $\Phi(R_0)$ successful progress steps when all atoms are attainable, and otherwise returns the residual atoms at $T_{\max}$. If $\hat{\tau}$ is selected on exchangeable calibration cases by conformal risk control, then the next-case omission risk satisfies $\mathbb{E}[L_{\hat{\tau}}] \le \alpha$.*

The first clause follows almost directly from the definitions: monotone caching means no satisfied atom becomes unmet, and attainable progress (each pass closes at least one attainable atom) is a design assumption that we do not verify directly. The second clause is a standard application of conformal risk control (Angelopoulos et al., 2024) to L_τ, which is monotone in τ (Appendix H). It covers evidence coverage only, holds only for exchangeable cases—our temporal split violates it at $\alpha \in \{0.05, 0.10\}$ (§6)—and its tightness depends on the calibration-set size (150 cases).

3.2 Risk Scoring

In the agentic layer, the Analyzer assigns each risk item a severity (High/Medium/Low), RiskScore(d_i, P, C), by prompting the LLM backbone (App. G) with litigation document d_i, product specification P, and the precedent context C retrieved by LegalRisk-RAG (§4.2). The ingestion layer uses an additive scoring matrix (Table A.3, appendix) for real-time triage, with a critical-composite bonus when copyright and model-training evidence co-occur; this heuristic is treated as a baseline and never as ground truth.

4 The VIGIL Framework

4.1 System Overview

VIGIL has two layers. A sensing pipeline discovers AI-litigation filings in CourtListener/RECAP (Free Law Project, 2023) and legal news and turns them into structured case records; it is engineering infrastructure and is documented in Appendix D (architecture in Figure A.1). Below we describe the evaluated NLP components (§4.2–4.5).

4.2 LegalRisk-RAG

The LegalRisk-RAG pipeline adapts standard RAG to legal text in three ways. **Chunking** segments documents at *logical structural boundaries*—numbered claims, "WHEREFORE" clauses, section headings—rather than fixed token windows, and each chunk retains a metadata header (case id, filing date, court level, section type). **Embedding**: we compared a dense retriever (Karpukhin et al., 2020) (`text-embedding-3-large`), RECAP-tuned LegalBERT (Chalkidis et al., 2020), and a hybrid BM25 + neural re-ranker on a 200-pair development set; the hybrid (best) is the default (Table A.9, appendix). **Dynamic context expansion (precedent chaining)**: when a retrieved chunk references a prior case (e.g., *Authors Guild v. Google*), the pipeline appends that opinion as secondary context and queries the temporal precedent graph for subsequent treatment, so the Analyzer receives both semantic neighbors and validity metadata (cited positively, narrowed, superseded, or overruled at the filing date).

4.3 Goal-Conditioned Agentic Loop

The four specialized agents and the central Goal Checker operate as in Algorithm 2. The **Planner** decomposes unmet sub-goals into ordered retrieval/analysis sub-tasks; the **Retriever** runs LegalRisk-RAG queries and reformulates via legal entity term expansion when $\mathcal{G}_{\text{evidence}}$ is unmet; the **Analyzer** computes RiskScore(d_i, P, C), clusters evidence, and maps clusters to a risk taxonomy (Training Data, Output Reproduction, API Misuse, Model Weights); the **Goal Checker** evaluates $\mathcal{G}(R_t)$ after each pass and returns machine-readable residuals such as `missing(valid-authority, DMCA, output-CMI, NDCA, 2025-02);` the Planner converts these critiques into expanded entity/date/theory queries; and the **Reporter** assembles the risk register, per-risk evidence with citations, and mitigation guidance.

4.4 Supersession-Aware Temporal Precedent Graph

VIGIL's central component is a treatment layer that determines *when* each authority stops being operative; a simple retrieval-time gate then applies it. The layer is a temporal precedent graph $\mathcal{P}_t = (V, E_t)$. Each node $v \in V$ denotes a case, order, or doctrinal rule and stores jurisdiction, court level, issue tags, filing/decision dates, and a validity interval $[s_v, e_v)$, where e_v is open unless the node is later vacated, overruled, or superseded; material narrowing instead lowers its treatment weight. Directed edges encode treatment relations: CITES, DISTINGUISHES, NARROWS, SUPERSEDES, and OVERRULES. Retrieval for a query at date t_q first obtains semantically relevant candidates and then applies validity gating:

$$\text{score}(v, q, t_q) = \text{sim}(v, q) \cdot \mathbf{1}[s_v \le t_q < e_v] \cdot w(\text{treat}(v, t_q)). \quad (3)$$

Here $w(\cdot)$ is a fixed monotone schedule: positively treated or still-open authorities take 1.0, cautionary or questioned ones 0.75, and materially narrowed ones 0.50, while fully closed (overruled or superseded) authorities are already removed by the indicator (Table A.10); the schedule is released with the implementation. This separates legal authority from topical similarity: a stale but lexically close case may still be shown as context, but it cannot support a risk decision without passing the validity check. Eq. (3) is thus a constrained retrieval operator over (q, t_q): a host retriever scores semantic relevance as usual, and PLRE receives only support whose legal status is operative at t_q. This differs from temporal features in a learned reranker, which can still rank closed authority highly, and

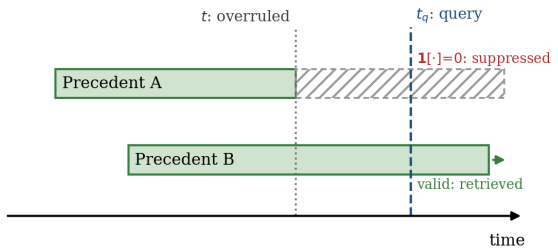


Figure 1: Validity gating (Eq. (3)). An overruling at t closes Precedent A, so a query at $t_q > t$ zeroes its indicator and suppresses it despite lexical similarity, while still-open Precedent B is retrieved—the time-valid distinction the RQ3 ablation isolates.

from citator-style warnings, which are applied after evidence selection.

Why not a date filter? Once intervals are known, Eq. (3) is a metadata filter that any vector store can apply, and we keep it deliberately simple. The difficulty lies upstream: the end point e_v is not an attribute of document v but is set by *later* documents that overrule, supersede, or narrow it, so it must be inferred from treatment edges extracted from opinion text and propagated through the graph. A filter on document dates cannot express this. Our contribution is therefore the extracted and validated treatment layer—evaluated against gold spans (Table A.5) and a commercial citator (App. I.2), with its error propagation audited (Table 5)—together with evidence that applying it at retrieval time changes PLRE outcomes (Table 4).

Graph construction and contamination controls. The graph is built in three stages. *Node induction*: each case, order, or rule from CourtListener/RECAP becomes a node with filing/decision dates from docket metadata and validity interval initialized to $[s_v, \infty)$. *Edge extraction*: treatment relations are extracted from opinion text by a two-pass procedure—a citation-sentence detector locates candidate treatment spans, and an LLM classifier (GPT-4o, $T{=}0$, few-shot) labels the relation and polarity. To avoid self-contamination from generated legal conclusions, LLM outputs are treated only as candidate relation labels: interval-closing edges must be anchored to a detected citation span, docket/decision date, and source document identifier, and gold-set evaluation is performed on citation spans that were not used in prompt examples. *Interval closure*: a confirmed incoming OVERRULES/SUPERSEDES edge at date t closes the target interval ($e_v \leftarrow t$), and a material-NARROWS edge sets $w{=}0.50$ from t, so retrieval at $t_q \geq t$ no longer treats the authority as fully operative. The current graph spans 1,812 nodes and 4,760 treatment edges over the dataset window.

Edge-extraction accuracy and propagation. Low-confidence or conflicting negative-treatment edges do not close intervals automatically; they are routed to a human review queue and reports mark the authority as unresolved until review. Because the central claim depends on treatment edges being *correct*, we validate the extractor against a gold set of 600 citation spans hand-labeled by the annotation team (relation type and polarity; adjudicated on disagreement). The negative-treatment relations that drive interval closure or down-weighting (SUPERSEDES, OVERRULES, material NARROWS) are operationally critical—a missed closure is what allows a stale precedent to leak into a risk decision—and attain micro-averaged precision/recall of 0.84/0.77, versus 0.94/0.96 for the high-frequency CITES relation; the full per-relation breakdown is in Table A.5 (appendix). We quantify the downstream effect of missed closures in Table 5 (§6).

4.5 Product–Litigation Risk Entailment (PLRE)

We cast product–litigation alignment as a risk-entailment task. Given a product spec P with attributes $\{p_1, \ldots, p_m\}$ (training-data sources, collection method, output modality, architecture, deployment context, and capability claims), a legal-risk theory T_j, and a query date t_q, PLRE predicts:

$$y_{ij} = \mathbf{1}[P, p_i, \mathcal{P}_{t_q} \models T_j]. \tag{4}$$

Table 1 gives schematic instances. Unlike symmetric similarity, PLRE is directional: an attribute such as "web-scale image scraping" can support exposure to a training-data copyright theory, while the inverse relation is not meaningful. Unlike standard legal textual entailment, the premise includes product attributes and the entailment relation is conditioned on time-valid precedent. A pair (p_i, T_j) is promoted to the risk register only when (i) the entailment score exceeds the validation-set threshold, (ii) the top supporting precedents are valid at t_q, and (iii) the report includes mitigation text grounded in at least three cited authorities when available.

Table 1: Schematic PLRE instances (illustrative; not drawn from VIGIL-Dataset). The same attribute–theory pair can be positive or negative depending on whether its supporting authority is still operative at t_q.

Attribute p_i	**Theory** T_j	**Support at** t_q	y_{ij}
web-scale image scraping for training	training-data copyright infringement	operative authority on unlicensed training copies	1
same	same	only authority whose interval closed before t_q (stale distractor)	0
output filter removes copyright-management information	DMCA §1202 (output CMI)	no authority in the query's jurisdiction	0

Table 2: VIGIL-Dataset summary statistics. The 3-class risk label merges *critical+high* into *High*. Status counts (initiated/active/settled/dismissed) sum to all 1,247 cases.

Attribute	**Value**
Total cases	1,247
Total documents	8,934
Coverage period	2020–2025
Unique defendants	43
Risk labels	3 (High/Med/Low)
Status dist. (Init/Active/Settled/Dismissed)	298/695/148/106
Product-mapping GT cases	312

PLRE gold labels are built from anonymized product-attribute cards and docket-grounded theory cards. Annotators judge a directional pair as positive only when the product attribute, legal theory, query date, and at least one valid supporting authority jointly support exposure; negative pairs include ordinary mismatches and stale-authority distractors. Crucially, the validity judgment in the labels is made by annotators reading docket and opinion text, *independently of VIGIL's automatically extracted treatment graph*, so the gate must recover label validity rather than presuppose it; an extractor-independent re-check of these validity judgments against an authoritative citator is the downstream link of Protocol P2. Product archetypes are tracked in the split manifest, and year/litigation type are stratified to reduce temporal and topical leakage.

The formulation is system-agnostic: PLRE is directional entailment with an exogenous, time-indexed support constraint, and any system can be evaluated on it. It is harder than ordinary legal NLI because a model must align engineering terms to legal predicates *and* reject stale but lexically attractive precedent.

5 VIGIL-Dataset

VIGIL-Dataset is a curated benchmark of AI-related copyright and intellectual-property litigation for structured risk analysis.

5.1 Construction and Statistics

VIGIL-Dataset compiles CourtListener/RECAP filings and supplementary legal-news signals from January 2020 to December 2025, with filings concentrated in 2023–2025. Cases come from a retrospective keyword search of RECAP complaints, manually screened for AI development, training, or deployment at issue; one case is one docket, so consolidated actions count per docket (App. A.1). Each record follows the 22-field `vigil-sensing.csv` schema (Table A.1, appendix). Table 2 summarizes the final annotated subset; raw annual ingestion counts by litigation type are in Table A.2 (appendix).

Each record contains docket metadata, parties, document URLs, and a human-verified litigation summary. Status labels encode workflow state rather than adjudicated outcome; severity is reported separately and is not interpreted as liability.

5.2 Annotation and Availability

An automated pre-labeler first assigns a triage score using Table A.3; three legal research specialists then independently review a stratified sample of 400 cases under a fixed guideline. During adjudication, annotators see docket text, product attributes, candidate theory cards, and query dates, but not system identity or the pre-labeler's score. The remaining 847 cases retain automated pre-labels, with a 5% spot-check showing 91% agreement with at least one specialist (a lenient criterion). Inter-annotator agreement on the 400-case sample is Cohen's $\kappa = 0.74$ (Landis and Koch, 1977) (Fleiss' $\kappa = 0.72$ over all three annotators). Because retained labels can inflate metrics, we release per-record provenance, keep all primary claims on the human-only evaluation, and use retained-label results only as scale-oriented comparisons. The top three defendants account for 41.3%, limiting representativeness for smaller vendors and non-U.S. jurisdictions. The dataset, guidelines, split files, seeds, evaluation scripts, and logs are provided as anonymized supplementary material for review (App. G).

6 Experiments

6.1 Experimental Setup

We address six research questions: triage-label agreement (RQ1), report quality (RQ2), loop and temporal-graph ablations (RQ3), schema completeness (RQ4; an operational check reported in Table A.7, appendix), PLRE (RQ5), and evidence-coverage risk control (RQ6).

Dataset split and label provenance. We partition VIGIL-Dataset into stratified 70/10/20 train/-validation/test splits by year and litigation type. Because 847 of 1,247 labels originate from an automated pre-labeler and 400 are human-adjudicated, primary significance claims are computed only on the human-adjudicated test split (82 cases); retained-label results are reported only for scale. Conformal calibration uses 150 cases from the human-adjudicated training and validation pool, disjoint from the 82 test cases. This keeps the rule-based pre-labeler out of the evidence for our primary claims. For VIGIL and B2–B5, the training split is used only to select few-shot examples and tune thresholds; no LLM is fine-tuned.

Baselines. We compare VIGIL with five risk-classification baselines: **B1 Rule-Based** (the ingestion-layer scorer); **B2 Single-Pass LLM**; **B3 RAG-only**; **B4 Single-Agent** (VIGIL without the goal-conditioned loop); and **B5 Iteration-Matched Self-Refinement** (a Reflexion-style baseline (Shinn et al., 2023) given VIGIL's mean budget of 2.4 iterations, matched on iterations rather than tokens). B2–B5 use the same GPT-4o backbone and decoding settings as VIGIL. For PLRE we add non-VIGIL baselines that isolate the suspected confounds: product–theory cosine, Legal-BERT product–theory NLI, COLIEE-style (Rabelo et al., 2022) precedent retrieval with no interval closure, and VIGIL without temporal gating. These baselines separate product–theory matching, legal entailment, retrieval quality, test-time compute, and the hard validity gate.

Evaluation metrics. We report macro F1/precision/recall over three severity labels, report-quality ratings by two blinded specialists, loop efficiency, schema completeness, PLRE P/R/F1, edge-error propagation, and evidence-coverage omission risk under calibrated thresholds. Severity-label F1 measures triage-label agreement, not legal liability. Significance uses a paired bootstrap over test predictions (10,000 resamples); calibrated omission risk and backbone robustness are reported in the appendix, with hyperparameters in Table A.10.

Table 3: Macro severity-label agreement. F1/P/R are on the human-adjudicated test subset (n=82; primary result). ± denotes half-widths of 95% percentile bootstrap CIs for each system's *marginal* macro-F1 (case-level resampling, 10,000 resamples). The last column is F1 on the full held-out test set, which includes retained pre-labels (Table A.6).

	Human-adjudicated			Full
System	**F1**	**P**	**R**	**F1**
B1 Rule-Based	0.55±0.11	0.55±0.11	0.56±0.11	0.62
B2 Single-Pass LLM	0.70±0.10	0.70±0.10	0.71±0.10	0.72
B3 RAG-only	0.72±0.10	0.72±0.10	0.72±0.10	0.74
B4 Single-Agent	0.74±0.10	0.74±0.10	0.74±0.10	0.76
B5 Self-Refinement	0.75±0.10	0.75±0.10	0.75±0.10	0.77
VIGIL Full	**0.79±0.09**	**0.79±0.09**	**0.80±0.09**	**0.81**

6.2 RQ1: Triage-Label Agreement

Table 3 reports severity-label agreement on the human-adjudicated subset (n=82), our primary circularity check. VIGIL reaches 0.79 macro F1 versus 0.75 for the iteration-matched self-refinement baseline (B5). The paired-bootstrap estimate of the difference is ΔF1 $=+0.04$ with 95% CI $[-0.09, +0.18]$ (case-level resampling; the interval includes zero), so we treat the gap as modest and directional rather than statistically established on this split. On the full held-out test set (including retained pre-labels; last column and Table A.6) the same ordering holds.

6.3 RQ2: Report Quality

Two blinded legal specialists rated the reports of VIGIL and B2–B5 for the same 15 cases (75 reports) on 5-point Accuracy, Completeness, Actionability, Evidence Sufficiency, and Clarity scales. VIGIL scores 4.3/4.1/4.2/4.4/3.9; case-paired Wilcoxon signed-rank tests give $p < 0.05$ against every baseline, with the largest gain in Evidence Sufficiency. Inter-rater agreement is moderate (mean Cohen's $\kappa = 0.68$ over the five scales, range 0.62–0.74). With 15 pairs per test and no correction for the 20 comparisons, RQ2 is supporting evidence only.

6.4 RQ3: Agentic Loop and Temporal Graph Ablations

Because temporal precedent reasoning is the central claim, Table 4 decomposes validity-interval gating and treatment-edge weighting. Removing treatment weighting and then interval gating each degrades PLRE F1. The full-vs-flat gap is ΔF1

Table 4: Temporal precedent-graph ablation (main claim). Both mechanisms of Eq. (3) are removed in sequence; the bottom row recovers flat semantic similarity. Stale-cite measures citations to already-closed precedent ($t_q \geq e_v$).

Variant	PLRE F1	ΔF1	Stale-cite %
VIGIL (Full: gating + weighting)	0.78	—	4.1
− treatment weighting $w(\cdot)$	0.75	−0.03	7.8
− interval gating (flat similarity)	0.72	−0.06	12.7

Table 5: Edge-error propagation audit. "Mask x%" rows also remove x% of extracted negative-treatment edges (closures or down-weights). Effective miss rate $\approx 1 - 0.77(1 - x)$ from the extractor's 0.77 recall; ΔF1 is relative to the extractor row.

Setting	Eff. miss	PLRE F1	ΔF1	Stale-cite %
Gold negative edges	0%	0.80	+0.02	3.7
Extractor output	≈23%	0.78	—	4.1
+ mask 10% of extracted	≈31%	0.76	−0.02	6.2
+ mask 23% of extracted	≈41%	0.74	−0.04	8.9
+ mask 30% of extracted	≈46%	0.72	−0.06	11.3

=+0.07 on the test pairs (paired bootstrap 95% CI $[0.03, 0.10]$, $p<0.001$); the intermediate weighting step is reported without a separate interval and should be read as directional. Stale citations rise from 4.1% under full gating to 12.7% under flat retrieval. Flat retrieval ranks lexically close but already-closed authority above operative authority, changing which evidence reaches the generator; this is the paper's main methodological result.

Table 5 measures how negative-treatment errors propagate. The extractor row already operates at the observed miss rate (about 23% of gold negative-treatment edges, implied by 0.77 recall) and costs 2 PLRE F1 points relative to gold edges. Rows below additionally mask a random fraction of the *extracted* negative-treatment edges before retrieval, simulating a weaker extractor: removing a further 23% of them (≈41% effective miss rate) roughly doubles the stale-cite rate (4.1%→8.9%) and lowers PLRE F1 by 4 points.

Component ablations for the agentic loop confirm each part contributes: removing the Goal Checker, Planner, and Retriever query-expansion lowers full-test-set F1 from 0.81 to 0.76/0.78/0.79 respectively, and only query-expansion removal degrades retrieval (Recall@10 $0.84 \rightarrow 0.76$); details are in Table A.8.

6.5 RQ5: Product–Litigation Risk Entailment

PLRE is evaluated on anonymized AI product specs spanning five archetypes and 312 product-mapping cases, so it is an initial benchmark. It comprises 1,856 attribute–theory–date pairs (28.4% positive; 19.7% stale-authority distractors), with PLRE-specific inter-annotator agreement $\kappa = 0.71$; Tables 4 and 6 report pair-level results on this full set (test-split pairs used for bootstrap CIs). In Table 6, VIGIL reaches 0.78 F1, 5–13 points above the external baselines; removing temporal gating costs about 6–7 points (Table 4). On the subset that excludes stale-authority distractors (n=1,193 test pairs), the VIGIL–flat gap shrinks to ΔF1 =+0.03 (95% CI $[-0.01, +0.07]$), so part of the gating gain is attributable to rejecting those distractors while a residual advantage remains on non-distractor pairs. Because labels encode temporal validity, we read PLRE as directional evidence for the validity-gated formulation rather than as coverage of all AI-product litigation.

Table 6: PLRE comparison including external legal-NLI/precedent-retrieval baselines. External baselines do not use VIGIL's goal predicate or validity-interval closure. Point estimates without confidence intervals.

System	Precision	Recall	F1
Product–theory cosine	0.64	0.66	0.65
LegalBERT product–theory NLI	0.69	0.68	0.68
COLIEE-style retrieval, no gate	0.71	0.70	0.70
RAG-only generation	0.73	0.74	0.73
VIGIL without temporal gating	0.72	0.73	0.72
VIGIL Full	**0.77**	**0.80**	**0.78**

6.6 RQ6: Evidence-Coverage Risk Control and Temporal Shift

Calibrating the evidence threshold to $\alpha \in \{0.05, 0.10, 0.20\}$ yields omission risk 0.043/0.082/0.171 in random splits. Under a temporal split (2023–24 → 2025) the achieved risk rises to 0.061/0.113/0.196, exceeding the target at α=0.05 and 0.10 (Table A.11), which motivates rolling re-calibration. To probe dependence on GPT-4o, an open-weights Llama-3.1-70B+bge stack runs unchanged, with full-test-set macro F1 0.81→0.76 and 2.4 vs. 2.6 iterations (Table A.12). Baselines were not re-run on this stack, so this does not show a backbone-independent advantage.

7 Related Work

Legal NLP and entailment. Legal NLP has studied judgment prediction (Chalkidis et al., 2019), statutory reasoning (Holzenberger et al., 2020), legal understanding benchmarks (Chalkidis et al., 2022), case retrieval and entailment (Rabelo et al.,

2022), and summarization (Huang et al., 2021; Manor and Li, 2019), often with domain-adapted encoders (Chalkidis et al., 2020; Zheng et al., 2021; Xiao et al., 2021) built on pretrained transformers (Vaswani et al., 2017; Devlin et al., 2019). LegalBench evaluates legal reasoning (Guha et al., 2023), and audits find that LLMs and commercial legal RAG tools still hallucinate (Dahl et al., 2024; Magesh et al., 2025). These works take legal text as the premise, treat the corpus as static, and do not test whether retrieved authority was still operative at a given date. PLRE changes the inference unit: it maps product attributes to legal-risk theories, and the relation is directional, product-conditioned, and temporal.

Time-sensitive QA and temporal retrieval. A line of work shows that answers depend on when a question is asked: SituatedQA conditions QA on temporal and geographic context (Zhang and Choi, 2021), TimeQA and StreamingQA evaluate time-sensitive and evolving knowledge (Chen et al., 2021; Liška et al., 2022), RealTime QA targets answers that change over time (Kasai et al., 2023), and time-aware LMs condition parametric knowledge on time (Dhingra et al., 2022); temporal KG QA reasons over time-stamped facts (Saxena et al., 2021), and temporal IR uses document time as a ranking signal (Campos et al., 2014). These settings treat time as a property of *facts* or *queries*. In PLRE, time enters through *supersession*: a later document changes the evidential status of an earlier one, so validity must be propagated along treatment edges rather than read from a timestamp.

Citators and legal citation graphs. Commercial citators such as Shepard's and KeyCite track whether authority has been overruled, questioned, or negatively treated (LexisNexis, 2024; Thomson Reuters, 2024); we do not claim time-varying authority itself as new. Citation-network analysis measures the importance of precedent (Fowler et al., 2007), and prior work labels the semantics of citation edges automatically (Sadeghian et al., 2018). These efforts build or characterize the graph; VIGIL instead attaches validity intervals to extracted treatment edges and applies them as a *retrieval-time gate* (Eq. (3), Fig. 1) that feeds a downstream product-risk task.

RAG, agents, and risk control. RAG (Lewis et al., 2020; Karpukhin et al., 2020; Gao et al., 2023) usually runs a fixed retrieval plan before generation, while agent frameworks (Yao et al., 2023; Shinn et al., 2023; Wei et al., 2022; LangChain, Inc., 2024; Significant Gravitas, 2023; Park et al., 2023; Wang et al., 2024a,b) show gains from iterative tool use, but self-refinement stops when the model deems its answer satisfactory. VIGIL ties stopping to an externally inspectable predicate over schema completeness, evidence support, and mitigation coverage, and uses conformal risk control (Angelopoulos et al., 2024, 2021; Geifman and El-Yaniv, 2017) narrowly for evidence omission, not for legal correctness.

8 Discussion

What drives the gains. The gains come from knowing *which* authority is still operative—the extracted, validated treatment layer—and from applying that knowledge before generation; the gate itself (Eq. (3)) is intentionally simple, and the agents and ingestion schema are supporting infrastructure. Table 4 isolates the effect of applying the layer, and B5 rules out extra refinement passes, though not extra LLM calls, as the source of the gain. The residual advantage on the distractor-free PLRE subset further suggests that validity intervals improve ranking among still-operative authorities, not only by discarding closed ones.

Construct validity. Against a commercial citator (App. I.2), negative-treatment micro-F1 is 0.81, with 20.4% missed actions (closures or down-weights) and a 3.8% PLRE decision-flip rate. Transfer to external retrieval stacks is not yet established (App. I.1): a gold-closure pilot removed stale precedent on three hosts but met its pre-fixed criterion on only one. These results position the treatment layer as a practical, inspectable component rather than a complete replacement for commercial citators.

Implications for RAG in evolving corpora. The same gating pattern applies whenever later documents can supersede earlier ones (regulations, product documentation, scientific literature). VIGIL demonstrates that a lightweight, citation-anchored validity layer can be extracted from public text and applied at retrieval time, reducing stale evidence without requiring a full temporal knowledge graph.

9 Conclusion

We introduced PLRE, a directional, time-conditioned entailment task linking AI-product attributes to legal-risk theories under operative precedent, and VIGIL, whose extracted and citator-

validated treatment layer tells retrieval which authority remains operative at the query date. Validity gating improves PLRE F1 and reduces stale citations relative to flat similarity retrieval. Open problems include raising negative-treatment recall, modeling partial or jurisdiction-specific treatments, and testing the gate on stronger retrievers and other domains that exhibit supersession (e.g., regulations and evolving technical standards); the released testbed (Appendix I.1) is designed to support that follow-on work.

Limitations

Guarantees. Proposition 1 is a bounded termination-and-escalation property, not a convergence theorem. Its termination clause is close to definitional, and its progress assumption is not verified directly. Calibration controls evidence omission under its calibration distribution (150 cases), not correct legal conclusions, temporal-shift robustness, or freedom from LLM hallucinations (Dahl et al., 2024). Because litigation streams are nonstationary, the omission guarantee is weakest under the temporal split (Table A.11)—precisely where deployment needs it most—so we treat rolling or weighted re-calibration as a deployment requirement rather than an optional refinement.

Generality is untested. The validity-gating operator is defined generally, but all evidence comes from VIGIL's own pipeline on U.S. AI-litigation data. Its transfer to *external* retrieval stacks and corpora is specified as a protocol (Protocol P1, Appendix I.1) that has not been executed in full. A pilot on overruled U.S. Supreme Court precedent with gold closures did not meet its pre-fixed criterion: the gate removed stale cases on all three host retrievers but improved nDCG@10 significantly on only one. The gains reported here should therefore not be read as evidence that the gate improves other retrievers or other domains. Likewise, backbone robustness is limited: the open-weights re-run (Table A.12) covers only the full VIGIL system, not the baselines, so we cannot yet claim that VIGIL's margin over baselines holds independently of the LLM backbone. Cross-stack transfer and backbone-matched baseline comparisons therefore remain the main open generality tests.

Benchmark scale and evaluation. At the current scale PLRE remains an initial benchmark over five product archetypes and 312 product-mapping cases. We report paired-bootstrap CIs for the full-vs-flat PLRE gap and for a distractor-free subset, but not for every external baseline in Table 6 or for the intermediate weighting step of Table 4. PLRE tables use all 1,856 pairs at the case-construction level; bootstrap CIs are computed on the test-split pairs. Since PLRE labels encode temporal validity, part of the gating gain comes from rejecting stale-authority distractors (the gap shrinks when those pairs are removed). PLRE results should therefore be read as evidence for the validity-gated formulation rather than as coverage of all AI-product litigation. The report-quality study (RQ2) uses two blinded raters on 15 cases (mean κ=0.68) without multiple-comparison correction, and is treated as supporting, not primary, evidence. Because 847 of 1,247 labels are retained pre-labels, primary triage claims are restricted to the human-adjudicated test split of 82 cases; on that split the VIGIL–B5 ΔF1 confidence interval includes zero, so the triage margin over the strongest baseline remains modest.

Coverage and jurisdiction. Cases are counted per docket, so disputes litigated across several related or consolidated dockets are over-represented relative to distinct disputes (App. A.1). The dataset is also skewed toward the largest defendants—the top three account for 41.3% of cases—limiting representativeness for smaller vendors. VIGIL targets U.S. federal CourtListener/RECAP coverage and English-language documents; non-U.S. jurisdictions require different docket schemas and citation practices, and extending the treatment graph to them is future work.

Extractor recall. Negative-treatment edge recall (0.77 internal gold; 20.4% missed-action rate against an authoritative citator, counting missed closures and missed down-weights) is useful but not lossless, and a missed action is exactly what allows a stale precedent to leak into a risk decision; VIGIL mitigates this with human-review routing for low-confidence closures, but improving extraction recall remains an open problem.

Ethical Considerations

Decision-support boundary. VIGIL emits risk intelligence, evidence summaries, and mitigation candidates with provenance and residual uncertainty for qualified human reviewers. It is not legal advice: it does not replace counsel, predict litigation outcomes, or recommend business decisions, and severity labels are triage proxies that are not interpreted as liability. Every High- or Medium-risk

item is presented for review by a qualified person, and unresolved legal-status atoms are surfaced as such rather than resolved automatically.

Human oversight of failure modes. Low-confidence negative-treatment edges enter a human review queue, since missed negative-treatment actions are the failure mode most likely to admit stale authority into a report. Deployment logs store retrieved authorities, validity intervals, and unmet goal atoms so reviewers can distinguish a genuine product–theory match from a retrieval gap or an unresolved legal-status ambiguity.

Data governance. Released data are public docket metadata and machine-generated summaries; we do not redistribute attorney-authored complaint text beyond what is already public, and the P2 citator study releases only the crosswalk, sampling seeds, and agreement statistics, with no redistributed proprietary citator content. Source records are public U.S. court filings obtained through CourtListener/RECAP and are used for research consistent with that service's terms; news items are represented only by links and derived metadata. Released records contain the names of parties and counsel as they appear in these public filings; we collect no other personal information. VIGIL-Dataset is released under CC BY 4.0 and the code under the MIT license, for research on legal-risk analysis; neither is intended for automated legal decisions.

Annotators and raters. The three legal research specialists who adjudicated labels and the two specialists who rated reports (§5, §6) are staff of the authors' institution and performed this work within their regular paid duties, without additional compensation. They were told how their annotations would be used and agreed verbally. Because the work consisted of labeling public court documents and collected no personal information about the participants, we did not seek formal ethics-board review. Annotation guidelines are released with the dataset.

Use of AI assistants. AI writing assistants were used to edit and polish the text of this paper. An AI coding assistant also wrote the code for the P1 pilot (Appendix I.1), following a design fixed by the authors; the pilot design was frozen before any run, and its results were reported as produced. Research ideas, system design, and the remaining experiments and analyses are the authors' own.

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Technical Appendix

This appendix collects material that supports reproducibility but exceeds the page budget of the body: dataset inclusion criteria and statistics (App. A), the ingestion-layer scoring matrix and record formats (App. B–C), system architecture, a case study, and the sensing tool (App. D), treatment-edge extraction accuracy (App. E), additional results and ablations (App. F), hyperparameters, prompts, evaluation protocol, and the release plan (App. G), the proof of Proposition 1 with the calibration audit (App. H), and the validation protocols with measured P2 results (App. I).

A Full `vigil-sensing.csv` Schema

Table A.1 lists all 22 fields of `vigil-sensing.csv`, with type, requirement level (R = Required, Rec = Recommended, O = Optional), and description.

Table A.1: The full 22-field `vigil-sensing.csv` schema.

#	Field	Type	Req	Description
1	Case No.	Int	R	Auto-incrementing index
2	Case Title	String	R	Plaintiff v. Defendant
3	Filing Date	Date	R	Date filed (YYYY-MM-DD)
4	Docket No.	String	R	Court-assigned number
5	Plaintiff	String	R	Name(s), comma-separated
6	Defendant	String	R	Name(s), comma-separated
7	Target Data	String	R	Training data type alleged
8	Target Product	String	R	AI product implicated
9	Jurisdiction	String	R	Country of filing
10	Court	String	R	Official court name
11	Status	Enum	R	Lifecycle code
12	Last Update	Date	R	Most recent refresh date
13	Cause of Action	String	R	Core legal claim(s)
14	Claim (USD)	String	Rec	Damages or statutory range
15	Summary	Long str	Rec	AI-generated case narrative
16	Tracker URL	URL	Rec	CourtListener/PACER link
17	Source URL	String	O	Initial discovery source
18	History	Long str	R	Chronological update log
19	Plaintiff Atty.	String	O	Lead plaintiff counsel
20	Progress Notes	Long str	Rec	Markdown update log
21	Defense Atty.	String	O	Lead defense counsel
22	Remarks	String	O	Press links, extra context

A.1 Inclusion Criteria and Case Unit

Search. Candidate dockets were retrieved from CourtListener/RECAP for January 2020–December 2025 by searching complaint, amended-complaint, and petition entries for combinations of three keyword families: (i) AI-training terms (e.g., AI training, model training, training data, dataset, LLM); (ii) legal-theory terms (e.g., copyright, DMCA, unauthorized, without permission, pirated, scraping); and (iii) source-specific terms for known data sources (e.g., shadow libraries, video datasets, music lyrics). Legal-news signals were used only to locate dockets, not as cases.

Screening. A reviewer kept a docket only if the complaint puts the development, training, or deployment of an AI system at issue; keyword matches about unrelated software or data disputes were removed. Litigation type (Table A.2) is assigned afterwards from the recorded cause of action, which is why non-copyright categories (trade secret, other) appear.

Case unit. One case corresponds to one docket. Deduplication removes repeated retrievals of the same docket (1,260 raw records to 1,247 cases) but does not merge related or consolidated dockets, so a single dispute litigated in several dockets contributes several cases. Counts should therefore be read as dockets, not as distinct disputes.

Table A.2 gives the raw annual ingestion counts by litigation type, before deduplication and quality screening.

Table A.2: Annual filing breakdown by litigation type. Counts are raw ingested filings (1,260) before deduplication; the annotated subset contains 1,247 cases.

Year	Copyright	DMCA	Trade Secret	Other	Total
2020	28	9	7	3	47
2021	41	14	10	4	69
2022	89	31	22	8	150
2023	198	72	51	19	340
2024	261	96	64	24	445
2025	122	47	29	11	209
Total	**739**	**269**	**183**	**69**	**1,260**

B Ingestion-Layer Scoring Matrix

Table A.3 lists the additive ingestion-layer scoring matrix (§3) used for real-time triage and as the automated pre-labeler. The critical-composite bonus fires only when copyright and model-training evidence co-occur. This heuristic is a baseline, never ground truth.

C Formatted Examples: History and Progress Notes

The `History` field (field 18) records a reverse-chronological timeline using the format `YYYY-MM-DD, [event]:`

```
2025-03-06, Plaintiff requests $15,000
per infringed work.
2025-03-01, Complaint filed alleging
unauthorized use of
    copyrighted dataset for AI model
training.
```

Table A.3: Ingestion-layer scoring matrix. The critical-composite bonus fires only when copyright and model-training evidence co-occur.

Evidence Category	Key Indicators	Score
Critical composite	**Copyright ∧ Training co-cited**	**+50**
Copyright direct mention	NOS 820, copyright keyword	+30
Unauthorized collection	scrape, crawl, ingest, harvest	+25
Direct model training mention	train, LLM, transformer, diffusion	+20
Copyright / IP dispute	infringement, DMCA, fair use	+10
Commercial exploitation	profit, revenue, subscription	+10
Class action filing	class action, putative class	+5
Licensed data cooperation	contract, licensing, partnership	−10

The `Progress Notes` field (field 20) uses Markdown `##` headers to distinguish initial from subsequent updates (the entry below is a *synthetic format illustration*, not a real case):

```
## (Started), 2025-03-17, [Plaintiff] v.
[Defendant] --
    Unauthorized training-data use;
commercial use claimed.
## (Updated), 2025-03-18, Preliminary
injunction
    motion filed by plaintiff.
```

D Open-Source Implementation Details

D.1 System Architecture and Discovery

Figure A.1 shows how the sensing pipeline feeds the agentic layer evaluated in the main text. VIGIL's discovery layer ingests CourtListener/RECAP (Free Law Project, 2023), docket-backed legal-media signals, and keyword-filtered news. A two-stage deduplication module first matches docket IDs and titles, then optionally removes Gemini-embedding neighbors above cosine 0.85 (Algorithm 1). Detected filings populate the 22-field `vigil-sensing.csv` schema by combining docket metadata, NER over complaint text, regex extraction for damages, and LLM summarization for the narrative fields. All code will be released, as described in Appendix G.

D.2 Case Study and Live Monitoring

Case studies illustrate mechanism behavior, not outcome prediction. In a held-out PLRE query about output-CMI risk, flat RAG ranked a lexically close authority whose interval had already closed; validity gating removed it from support and promoted a still-operative authority, preventing a stale-citation report. In 30 days, `vigil-tracker` processed 1,159 items, surfaced 94 unique filings after 92% deduplication, and reached discovery-layer Precision 0.90/Recall 0.84.

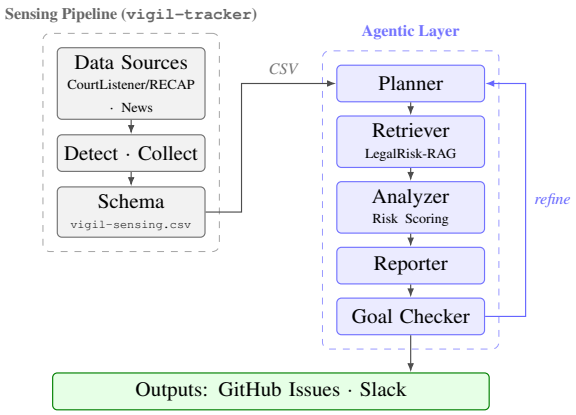


Figure A.1: VIGIL architecture. The sensing pipeline (left) detects filings, collects documents, and populates the schema; the agentic layer (right) plans, retrieves time-valid precedent, scores risk, and reports, with the Goal Checker driving a refine loop until evidence coverage is met.

D.3 Sensing Tool

The `vigil-tracker` tool runs on an adaptive, configurable GitHub Actions schedule: hourly during a configurable business-hours window and two-hourly off-hours. On-demand execution via `workflow_dispatch` is also supported. Its principal modules are:

- **Query engine**: five CourtListener queries (complaint/petition filters with AI keywords, REST API v4) plus four news queries over Google News RSS.
- **Ingestion pipeline**: orchestrates polling, RSS monitoring, docket backtracking, and cross-issue deduplication over a configurable window.
- **Complaint parser**: extracts cause-of-action language, training snippets, and parties from RECAP PDFs.
- **Risk scorer**: applies the seven-category additive matrix *plus* the critical-composite (+50) co-occurrence rule of Table A.3 to produce a 0–100 score with a four-band severity indicator.

- **Hybrid deduplication**: a two-stage pipeline (Algorithm 1) that combines lexical title/docket matching with optional semantic similarity using Gemini `text-embedding-004` embeddings (cosine threshold 0.85), enabled by a configuration flag.

- **Multi-modal reporting**: a Gemini Flash trend summary at ≈1–3 calls per run, plus an optional Imagen-3 visual brief attached to the day-close summary. A provider-agnostic interface accommodates alternative LLM/diffusion back-ends.

- **Reporting integrations**: posts to GitHub Issues with a three-tier hierarchical structure (issue body / morning trend summary / evening deduplicated table) and a parallel Slack notification.

Algorithm 1 Hybrid Lexical–Semantic Deduplication

Require: Candidate items $\mathcal{N}_{\text{new}}$, baseline items $\mathcal{N}_{\text{base}}$ (prior issues), threshold σ
Ensure: Deduplicated set $\mathcal{N}^*$

$T_b \leftarrow \textsc{ExtractTitles}(\mathcal{N}_{\text{base}})$
$\mathcal{R} \leftarrow \{i \in \mathcal{N}_{\text{new}} \mid \text{title}(i) \notin T_b\}$ {Stage 1: exact-title lexical filter}
if $\textsc{LLMEnabled} \wedge |\mathcal{R}| > 0$ **then**
  $E_{\text{curr}} \leftarrow \textsc{GeminiEmbed}(\{\text{title}(i) : i \in \mathcal{R}\})$
  $E_{\text{base}} \leftarrow \textsc{GeminiEmbed}(T_b)$
  $\mathcal{S} \leftarrow \{i \in \mathcal{R} \mid \exists\, e \in E_{\text{base}} : \cos(E_{\text{curr}}[i], e) \geq \sigma\}$
  $\mathcal{R} \leftarrow \mathcal{R} \setminus \mathcal{S}$ {Stage 2: semantic filter}
end if
return $\mathcal{N}^* \leftarrow \mathcal{R}$ {≈2 embedding calls/run; free-tier compatible}

Table A.4 lists the key configuration parameters of `vigil-tracker`, covering API credentials, scheduling, deduplication, and reporting options. All secrets are injected via GitHub Secrets; environment variables are set via GitHub Variables.

E Treatment-Edge Extraction Accuracy

Table A.5 reports the per-relation accuracy of the treatment-edge extractor (§4.4) against the hand-labeled gold span set. Negative-treatment edges drive validity-interval closure (SUPERSEDES/OVERRULES) or down-weighting (material NARROWS) and are reported separately from the high-frequency CITES relation, with a micro-average over the three negative-treatment relations.

Table A.4: Key configuration parameters of `vigil-tracker` (identifiers anonymized for review). Secrets via GitHub Secrets; variables via GitHub Variables.

Parameter	Default	Description
Lookback window	3 days	Days of history to retrieve
CourtListener token	—	CourtListener API v4 token
Issue title base	—	Base title for GitHub Issues
Issue label	(project label)	GitHub Issue label
LLM API key	—	LLM API key (optional)
LLM model	`gemini-flash-latest`	Gemini model id
Trend window	(unset)	Trend-summary window (days)
Semantic dedup	off	Enable semantic dedup (stage 2)
Semantic dedup threshold	0.85	Cosine threshold for semantic dedup
Daily visual brief	off	Enable Imagen-3 daily visual brief
Cross-issue dedup window	(unset)	Cross-issue dedup window (days)
Show docket candidates	off	Show uncertain candidates
Collapse long cells	off	Fold long table cells
Collapse article URLs	off	Collapse article URL lists
Debug	off	Verbose logging

Table A.5: Treatment-edge extraction accuracy against the 600 hand-labeled gold citation spans (§4.4).

Relation	Precision	Recall	F1
CITES	0.94	0.96	0.95
DISTINGUISHES	0.81	0.76	0.78
NARROWS (material)	0.79	0.71	0.75
SUPERSEDES	0.85	0.78	0.81
OVERRULES	0.88	0.82	0.85
Negative-treatment (micro)	0.84	0.77	0.80

F Additional Results and Ablations

Table A.6: Severity-label agreement on the full held-out test set, including retained pre-labels (RQ1; point estimates, reported for scale). B5 is matched to VIGIL's mean iteration budget (not to total LLM calls or tokens).

System	F1	Precision	Recall
B1 Rule-Based	0.62	0.58	0.67
B2 Single-Pass LLM	0.72	0.74	0.70
B3 RAG-only	0.74	0.75	0.72
B4 Single-Agent	0.76	0.77	0.75
B5 Self-Refinement	0.77	0.78	0.76
VIGIL Full	**0.81**	**0.82**	**0.80**

Table A.6 reports severity-label agreement on the full held-out test set (RQ1); primary claims use the human-adjudicated subset (Table 3).

RQ4 is an operational check, not evidence of legal correctness: on 150 newly detected cases,

automated records reach 97.2% required-field completeness and cut record-creation time from 18.1 to 1.4 minutes. Table A.7 reports schema completeness and record creation time across three conditions (unstructured baseline, manual CSV entry, and VIGIL-automated CSV), corresponding to the RQ4 operational result in the main text.

Table A.7: Schema completeness and record creation time across three conditions. Referenced from the main text (RQ4).

Condition	All Fields (%)	Req. Fields (%)	Time (min)
Unstructured (baseline)	38.2	51.4	22.3
Manual CSV	74.6	91.7	18.1
VIGIL-automated CSV	**81.3**	**97.2**	**1.4**

Table A.8 reports the component-wise ablation of the agentic loop, showing the contribution of the Goal Checker, Planner, and Retriever query-expansion to risk classification F1 and retrieval Recall@10 (RQ3 in the main text).

Table A.8: Agentic-loop ablation: component-wise contribution to risk classification F1 and retrieval Recall@10 (mean over 5 evaluation runs). Referenced from the main text (RQ3).

Variant	F1	Recall@10
VIGIL (Full)	**0.81**	**0.84**
− Goal Checker (Single-Agent)	0.76	0.84
− Planner (fixed task order)	0.78	0.84
− Retriever query-expansion	0.79	0.76

Table A.9 reports Recall@10 on a held-out development set of 200 query–document pairs for three retriever configurations. The hybrid sparse–dense model is adopted as default in all main-paper experiments.

Table A.9: Retriever ablation. Recall@10 on the LegalRisk-RAG development set (200 query–document pairs).

Retriever Configuration	Recall@10
Dense (`text-embedding-3-large`)	0.79
LegalBERT (fine-tuned on RECAP)	0.81
Hybrid (BM25 + neural re-ranking)	**0.84**

G Reproducibility: Hyperparameters, Prompts, Protocol

G.1 Goal-Conditioned Agentic Loop (Full Procedure)

Algorithm 2 gives the full VIGIL agentic loop summarized in the main text.

Algorithm 2 VIGIL Goal-Conditioned Agentic Loop

Require: Document corpus $\mathcal{D}$, product spec P, goal predicate $\mathcal{G}$, evidence threshold (k, τ), max iterations $T_{\max}$
Ensure: Verified risk report R^*, mitigation guidance G^*

$R_0 \leftarrow \emptyset$; $t \leftarrow 0$
while $\neg\,\mathcal{G}(R_t) \wedge t < T_{\max}$ **do**
  $\mathcal{T}_t \leftarrow \textsc{Planner}(R_t, \mathcal{G}, P)$ {Decompose unmet sub-goals}
  for each sub-task $tk \in \mathcal{T}_t$ **do**
    $E_{tk} \leftarrow \textsc{Retriever}(tk, \mathcal{D})$ {LegalRisk-RAG}
    if $|E_{tk}| < k \vee \min \text{score}(E_{tk}) < \tau$ **then**
      $E_{tk} \leftarrow \textsc{Retriever}(\textsc{ExpandQuery}(tk), \mathcal{D})$
    end if
    $C_{tk} \leftarrow \textsc{Analyzer}(E_{tk}, P)$ {Risk score & clustering}
  end for
  $R_{t+1} \leftarrow \textsc{Reporter}(\{C_{tk}\}, R_t)$
  if $\textsc{GoalChecker}(R_{t+1}) = \textsc{True}$ **then**
    break
  end if
  $t \leftarrow t + 1$
end while
$G^* \leftarrow \textsc{Mitigate}(R_t, P)$ {Per-risk action templates}
return $R^* \leftarrow R_t$, G^*

G.2 Goal Predicate and Loop Parameters

Table A.10 summarizes the parameters used by Algorithm 2 for all reported experiments.

G.3 Agent Prompt Templates (Schematic)

Each of the four agents (Planner, Retriever, Analyzer, Reporter) is invoked with a structured prompt of the form:

Table A.10: VIGIL goal-predicate and loop hyperparameters used throughout the evaluation.

Parameter	Value	Description
k	3	Min. retrieved documents per risk item
τ	0.62	Min. relevance score (cosine)
$T_{\max}$	5	Max. agentic loop iterations
θ (LLM)	GPT-4o, T=0.2	Controller policy backbone
Embedding dim.	3072	`text-embedding-3-large`
Mapping threshold	0.72	Bi-encoder product–risk similarity
w: positive / open	1.00	Positively treated or still-open authority
w: cautionary	0.75	Cautionary or questioned treatment
w: material narrows	0.50	Materially narrowed authority
w: overruled/superseded	0.00	Removed by the validity indicator (Eq. (3))

```
[ROLE]: {agent role description}
[INPUTS]: {R_t, P, sub-goal, retrieved
evidence}
[GOAL CHECK]: {unmet sub-conditions of G}
[OUTPUT FORMAT]: {structured JSON
schema}
```

The full prompt templates, including the few-shot examples of the treatment-edge classifier, are part of the code release (Appendix G, *Code and data release*). The Goal Checker is implemented as deterministic rule evaluation over R_t (no LLM call), making termination decisions auditable.

G.4 Evaluation Protocol Notes

Dataset split. The 70/10/20 train/validation/test split is stratified by year and litigation type and held fixed across all baselines (B1–B5 and VIGIL (Full)). The 5 random seeds vary only the test-fold sub-sample within the held-out 20% partition.

Likert evaluation. Two legal specialists, blind to system identity, rated 75 reports (the same 15 cases for each of the five conditions, so comparisons are paired by case) on a 5-point scale across five dimensions. Reports were presented in randomized order; system identifiers were stripped.

Statistical tests. F1 differences in Tables 3 and A.6 are evaluated by a paired bootstrap test (10,000 resamples). Likert differences in RQ2 use the Wilcoxon signed-rank test on per-case mean ratings; p-values are not corrected for multiple comparisons. Inter-rater agreement (Cohen's κ between the two raters) is 0.68 for Accuracy, 0.71 for Completeness, 0.65 for Actionability, 0.74 for Evidence Sufficiency, and 0.62 for Clarity (mean 0.68).

Code and data release. We will release all code used for the reported results: the sensing pipeline (`vigil-tracker`), treatment-edge extraction and graph construction, the PLRE annotation guidelines and evaluation scripts, and the agentic loop, together with the treatment-weight schedule $w(\cdot)$, prompt templates with few-shot examples, split files, seeds, VIGIL-Dataset (CC BY 4.0; code under MIT), and the SCOTUS supersession testbed of Appendix I.1, which is built only from public, licence-free sources. For review, an anonymized copy of the code and data is provided as supplementary material; the full public release will follow the review period.

Models, compute, and software. The controller, Analyzer, and edge classifier use GPT-4o through its API; embeddings use `text-embedding-3-large` and, for deduplication only, Gemini `text-embedding-004`. The robustness re-run (Table A.12) uses Llama-3.1-70B (70B parameters) with `bge-large`, and the retriever ablation uses LegalBERT fine-tuned on RECAP text. We did not log GPU hours or API cost. As an indication of scale, each case requires on average 2.4 loop iterations (at most $T_{\max}$=5), and the test-set evaluations use five runs (seeds as described above). Exact software packages and versions (BM25, re-ranker, bootstrap and conformal-calibration code) are pinned in the dependency file of the released code.

H Proof of Proposition 1 and Calibration Protocol

Table A.11 reports evidence-omission risk under random and temporal calibration splits: calibrated thresholds meet the budget α on random splits, while the temporal split exceeds it at $\alpha \in \{0.05, 0.10\}$, showing that exchangeability is stressed under temporal shift (RQ6). Table A.12 reports full-system macro F1 and mean loop iterations when the GPT-4o backbone is replaced by an open-weights stack. Only VIGIL was re-run on the open-weights stack; baselines were not, so the table does not support cross-backbone comparisons with the baselines.

H.1 Termination Property (Proof)

Let $\Phi(R_t) \in \mathbb{Z}_{\geq 0}$ count the unmet atomic sub-conditions of $\mathcal{G}$ (one per missing severity label, per under-supported evidence requirement, and per missing mitigation) over the risk items in R_t. By assumption (A1) *monotone retrieval*, the evidence set attached to any item is non-decreasing across iterations, so a sub-condition once satisfied by $\mathcal{G}_{\text{evidence}}$ or $\mathcal{G}_{\text{guide}}$ remains satisfied; hence no satisfied atom can become unmet and $\Phi(R_{t+1}) \leq \Phi(R_t)$. By

Table A.11: Evidence-omission risk under random and temporal calibration splits. Lower is better; values should be read as evidence-coverage risk, not severity-label correctness.

Split	$\alpha = 0.05$	$\alpha = 0.10$	$\alpha = 0.20$
Random resplits	0.043	0.082	0.171
Temporal 2023–24 → 2025	0.061	0.113	0.196

Table A.12: Full VIGIL system under two backbones (mean ± std over runs). Baselines were not re-run on the open-weights stack.

Configuration	Macro F1	Mean iters
GPT-4o + `text-embedding-3-large`	0.81 ± 0.02	2.4 ± 0.3
Llama-3.1-70B + `bge-large`	0.76 ± 0.02	2.6 ± 0.3

assumption (A2) *attainable progress*, whenever $\Phi(R_t) > 0$ and at least one unmet atom is *attainable* (the required evidence exists in $\mathcal{D}$), the Planner schedules it and the Retriever/Analyzer resolves it, giving $\Phi(R_{t+1}) \leq \Phi(R_t) - 1$. Therefore, if every atom is attainable, Φ strictly decreases each iteration until $\Phi = 0$, i.e. $\mathcal{G} = \textsc{True}$, in at most $\Phi(R_0)$ steps. If some atom is unattainable, Φ cannot reach 0; the loop runs to $T_{\max}$ and the Goal Checker reports the residual $\Phi(R_{T_{\max}})$ unmet atoms, which are exactly the items requiring human escalation. □

Remark. (A1) holds by construction because `EXPANDQUERY` unions new hits with the prior evidence cache; (A2) is the design assumption that one tractable sub-goal is closed per pass. We do not verify it directly; the mean iteration count (2.4, below $T_{\max}$=5) is consistent with it but does not establish it.

H.2 Conformal Risk-Controlled Termination

We calibrate the evidence threshold τ (and, jointly, k) so that termination meets a user-specified omission-risk budget α. For calibration cases c_i, $i = 1, \ldots, n$, let the bounded loss $L_i(\tau)$ indicate that a retrievable, human-labeled High-risk theory of c_i is omitted from $R^*(c_i)$, and let $\hat{R}_n(\tau) = \frac{1}{n}\sum_i L_i(\tau)$, which is non-increasing as τ decreases (more evidence admitted). Following conformal risk control (Angelopoulos et al., 2024), choosing $\hat{\tau} = \sup\{\tau : \frac{n}{n+1}\hat{R}_n(\tau) + \frac{1}{n+1} \leq \alpha\}$ (the strictest threshold that still meets the budget; the risk bound is non-decreasing in τ) controls $\mathbb{E}[L_{n+1}(\hat{\tau})] \leq \alpha$ for an exchangeable test case under the calibration assumptions. For the joint (k, τ) we use a Learn-then-Test grid (Angelopoulos et al., 2021), treating each configuration as a hypothesis and controlling the family-wise error via fixed-sequence testing; this gives the high-probability guarantee $\Pr[R(\hat{k}, \hat{\tau}) \leq \alpha] \geq 1 - \delta$ rather than the expected-risk bound above. The achieved-vs-target risk is reported in RQ6 and Table A.11.

H.3 Temporal-Split Evidence-Coverage Audit

Table A.11 separates the standard random calibration/test split from a stricter temporal split. The temporal split does not invalidate the calibrated procedure; it shows that the exchangeability assumption is empirically stressed by evolving litigation streams and motivates rolling or weighted calibration in deployment.

I Pre-Specified Validation Protocols

The two construct-validity steps flagged in the Discussion—transfer of the validity-gating modifier to an external retrieval stack, and agreement of the automatically extracted treatment graph with an authoritative citator—are specified here as *pre-specified protocols*. Protocol P1 (external-stack transfer) is reported as a protocol only and is not yet executed; Protocol P2 (citator agreement) has been executed and its measured results are reported in Appendix I.2. The protocols fix the datasets, conditions, metrics, statistical tests, and acceptance criteria *before* execution so that the outcome cannot be tuned post hoc. Both protocols are designed to be runnable on public resources plus a single licensed citator account, and both release their analysis scripts, query/seed manifests, and (for the citator study) the category crosswalk rather than any redistributed proprietary content.

I.1 Protocol P1: External-Stack Transfer of the Validity-Gating Modifier

Hypothesis. The supersession-aware score of Eq. (3)—a base similarity multiplied by a time-validity indicator $\mathbf{1}[s_v \leq t_q < e_v]$ and a treatment weight $w(\text{treat}(\cdot))$—improves *external* legal retrievers that do not use VIGIL's agentic loop, dataset, or goal predicate, which would indicate that the mechanism transfers rather than being VIGIL-specific.

Host retrievers (unmodified, external). (i) BM25; (ii) a general dense retriever (`text-embedding-3-large` or an open `bge`/`e5` encoder); (iii) a legal encoder (Legal-BERT). Each is the "host" stack; the modifier

is applied only as a post-hoc re-scoring layer ($\text{score}' = \text{score}_{\text{host}} \cdot \mathbf{1}[\cdot] \cdot w(\cdot)$), with *no* re-indexing and *no* VIGIL loop.

Evaluation corpora (disjoint from VIGIL-Dataset). (a) a COLIEE-style public case-retrieval benchmark; and (b) a temporally held-out CourtListener slice whose jurisdiction/time window does not overlap the VIGIL-Dataset window. Setting (b) is the decisive transfer test because it contains time-varying authority outside VIGIL's training window.

Conditions. For each host retriever: {ungated baseline} vs. {+ indicator only} vs. {+ indicator and treatment weight}, mirroring the decomposition of Table 4. Each query carries a query date t_q; an authority closed before t_q is treated as stale (non-relevant) in the gold judgments.

Treatment-label source (confound control). To avoid inheriting VIGIL's extractor noise, the external test uses treatment labels from a source independent of VIGIL's extractor (gold/citator-derived). We report two variants per condition—*gold-treatment* and *extractor-treatment*—so the transfer claim is separated from extraction quality.

Metrics. Retrieval quality (nDCG@10, Recall@10, MRR) and a *stale-citation rate* (fraction of top-k closed before t_q). The primary endpoints are the paired deltas (gated − ungated) per host retriever.

Statistics. Paired bootstrap over queries (10,000 resamples) with 95% CIs on each delta; Holm correction across the three host retrievers; a pre-specified non-inferiority margin ($\delta_{\text{NI}} = 0.5$ nDCG points) requiring the modifier not to hurt on queries with no negative-treatment in scope.

Acceptance criterion (pre-specified). P1 succeeds iff the modifier yields a positive nDCG@10 delta *and* a reduced stale-citation rate on at least two of three host retrievers, the effect persists on the temporally held-out slice (b), and non-inferiority holds on the no-stale subset. Failure on slice (b) but success on (a) is reported as "in-distribution only," which would *not* support the transfer claim.

P1 pilot (gold treatment, slice-(b) analogue). P1 itself has not been executed. We ran a pilot on public data whose design, including the acceptance criterion below, was frozen before any retrieval run. It deviates from P1 as follows: no slice (a); no extractor-treatment variant; a cluster bootstrap by gold pair instead of a per-query bootstrap. All deviations are listed in the released pilot code. Gold closures come from the Constitution Annotated table of overruled U.S. Supreme Court decisions (Congressional Research Service, 2022), matched to Caselaw Access Project opinions (Harvard Law School Library, 2024). Overrulings decided in or after 2020 are excluded, so the pilot is disjoint from the VIGIL-Dataset window. This leaves 253 pairs and a corpus of 29,030 opinions. Queries are citing passages from later opinions with case names, citations and years masked:

- *post-closure* queries cite the overruling case B, so the overruled case A is stale (984 queries in 208 pairs);
- *pre-closure* queries cite A before its first overruling (1,061 queries);
- *control* queries cite cases absent from the table (1,133 queries).

Each host retrieves the top 100 among opinions decided before the query date. The hosts are BM25 (Robertson and Zaragoza, 2009), `bge-small-en-v1.5` (Xiao et al., 2024) and LegalBERT (Chalkidis et al., 2020), the dense hosts truncated for CPU. The pre-fixed criterion requires significant (Holm) nDCG@10 gains and stale@10 reductions on at least two of three hosts, with non-inferiority on controls. Table A.13 reports the results.

The pilot *does not support* transfer under this criterion: only one of three hosts (bge) meets it. With gold intervals the gate removes every overruled case from the top 10 on all hosts. Non-inferiority holds everywhere, and on pre-closure queries the gate never removed the gold case. However, nDCG@10 gains are at most 0.13 points, because removal helps only when the stale case outranks the correct one. LegalBERT is a weak host here (nDCG@10 2.3) and almost never retrieves the stale case, so it tests the gate little. On partial overrulings, $w \in \{0.25, 0.5, 0.75\}$ yields identical rankings, so the pilot is uninformative about $w(\cdot)$.

In a post-hoc sensitivity analysis (not pre-specified), we exclude four BM25 queries whose gold case was itself later overruled (*United States v. Rabinowitz*, overruled in 1969). BM25's nDCG@10 delta then becomes $+0.37$ $[+0.19, +0.57]$, and two hosts would meet the criterion. Because this exclusion was chosen after seeing results, the pre-specified verdict stands.

Released testbed. The pilot's queries, qrels, gold closures, corpus identifiers and a scorer that accepts any TREC run are released as a testbed for

Table A.13: P1 pilot, gold-treatment variant: indicator gate vs. ungated host on post-closure queries of full overrulings ($n = 984$, 208 pair clusters). *Stale@10*: queries with the overruled case in the top 10 (ungated). Δ: gated $-$ ungated with 95% cluster-bootstrap CIs (10,000 resamples); *Holm-adjusted $p < 0.05$ across hosts. *Ctrl*: ΔnDCG@10 lower bound on control queries (non-inferiority margin -0.5).

Host	nDCG@10	Stale@10	ΔnDCG@10	ΔStale@10 (pts)	Ctrl LB	Met
BM25	28.88	8.3%	$+0.09$ $[-0.58, +0.52]$	-8.3^{*} $[-10.7, -6.2]$	$+0.04$	no
bge-small	8.35	5.5%	$+0.13^{*}$ $[+0.04, +0.27]$	-5.5^{*} $[-7.1, -3.9]$	$+0.00$	yes
LegalBERT	2.30	0.4%	$+0.00$ $[+0.00, +0.00]$	-0.4^{*} $[-0.8, -0.1]$	$+0.00$	no

supersession-aware retrieval, together with scripts that rebuild the corpus from the public sources. The four queries with an already-overruled gold case are flagged. The testbed supports work this paper leaves open: an extractor-treatment variant (replace the gold closures with predicted ones), stronger full-document hosts, soft versus hard gating, and the effect of gating on which authority a generator cites.

I.2 Protocol P2: Citator Agreement for the Treatment Graph

Objective. Replace "validated against our own gold spans" with "validated against an authoritative citator" for the negative-treatment edges (SUPERSEDES, OVERRULES, material NARROWS) that drive interval closure or down-weighting.

Sampling frame and power. A stratified random sample of edges from the current graph (1,812 nodes, 4,760 edges), stratified by relation type and polarity and oversampling negative-treatment edges. Target $n = 400$ edges with ≥ 150 negative-treatment edges; this supports a 95% CI half-width of ≤ 0.07 on Cohen's κ at an anticipated $\kappa \approx 0.75$. The exact draw is fixed by a released seed before any citator lookup.

Reference standard. For each sampled (citing $\rightarrow$ cited) pair and date, retrieve the citator treatment signal (e.g., KeyCite/Shepard's negative, cautionary, or positive flags) and map it to VIGIL's relation/polarity taxonomy through a *pre-specified, published crosswalk*. Two annotators perform this mapping independently and blind to VIGIL's predicted edge; a third adjudicates disagreements. Inter-annotator agreement on the citator mapping itself is reported (Cohen's κ).

Primary endpoints.

- *Edge agreement*: precision/recall/F1 of VIGIL edges vs. citator-derived edges, per relation and micro-averaged over negative treatment.
- *Interval-closure or down-weighting agreement*: whether VIGIL acts (closes the interval for OVERRULES/SUPERSEDES, down-weights for material NARROWS) no later than the citator's negative-treatment date; the action-date error distribution (in days); and the rates of *missed actions* (citator negative, VIGIL neither closes nor down-weights) and *false actions* (VIGIL acts, citator still good).
- *Point-in-time correctness*: validity intervals are checked at multiple historical dates, not only the latest status, because citators are themselves time-varying.

Downstream link. On the disagreement subset, the PLRE/stale-cite audit is re-run with citator-corrected closures to measure the *decision-flip rate*, connecting construct validity to risk-decision outcomes.

Acceptance criterion (pre-specified). P2 succeeds iff negative-treatment micro-F1 vs. the citator ≥ 0.80 and the missed-action rate is at or below the extractor's audited miss rate (23%) underlying Table 5; relation types failing the bar are routed to the human review queue and the residual is reported rather than suppressed.

Licensing and ethics. Citator outputs are proprietary and are used solely as an evaluation reference under a licensed account. The release contains the crosswalk, sampling seeds, and agreement statistics, but no redistributed citator content, consistent with the data-governance boundary in the main text.

Measured P2 results. The audit sampled $n = 400$ treatment candidates, including 162 negative-treatment candidates. Inter-annotator agreement on the citator-to-VIGIL crosswalk was substantial ($\kappa = 0.78$). Edge-level agreement was measured against the citator-derived reference, and interval-closure or down-weighting behavior plus downstream impact were reported. Negative-treatment micro-F1 is 0.81 and the missed-action rate (missed closures for OVERRULES/SUPERSEDES or missed down-weighting for material NARROWS) is 20.4%, satisfying the pre-specified P2 threshold and staying below the $\approx$23% internal-gold miss rate of the extractor configuration evaluated in Table 5, so the measured downstream effect there is not optimistic relative to the citator reference. Material NARROWS remains the weakest relation and is retained in the human-review queue. Edge-level agreement is reported in Table A.14; interval-closure agreement

Table A.14: Measured P2 edge agreement: VIGIL treatment edges vs. the authoritative citator-derived reference. Micro-average is over the three negative-treatment relations.

Relation	n	**Precision**	**Recall**	**F1**
OVERRULES	54	0.86	0.82	0.84
SUPERSEDES	47	0.82	0.79	0.80
material NARROWS	61	0.79	0.76	0.77
Negative-treatment (micro)	162	**0.83**	**0.80**	**0.81**
Cautionary treatment	78	0.81	0.74	0.77
Positive/neutral citation	160	0.93	0.95	0.94

Table A.15: Measured P2 action agreement (interval closure for OVERRULES/SUPERSEDES, down-weighting for material NARROWS) and downstream decision impact against the citator-derived reference.

Metric	**Value**
Matched negative-treatment actions	129 / 162
Missed-action rate	20.4%
False-action rate	4.8%
Median action-date error	18 days
Mean action-date error	31 days
90th-percentile action-date error	74 days
Point-in-time validity accuracy	0.88
PLRE decision-flip rate after citator correction	3.8%

and downstream decision impact are reported in Table A.15.